THE EAGLE FLASK COMPANY

Richard P. Rumelt

"It just can't be done." Mr. Fahimi, the managing director of the Eagle Manufacturing Company stated emphatically in May 1973. "We have concentrated for several years on changing our production methods to make our flasks more competitive. I think that all of the important steps in that direction have been taken by now but we are still faced with imports from the People's Republic of China with CIF prices that are less than our cost of raw materials. The government won't give us complete protection unless we raise our production level to the point where we can export, but we still wouldn't be able to beat the Chinese prices in the Gulf and neighboring states even if we tried to export. I don't see any future at all for the company and I have just about decided to leave it and look for a position elsewhere."

Located in Tehran, Iran, the Eagle Manufacturing Company was primarily a producer of a line of vacuum thermos flasks and Coleman thermos jugs. It also produced a variety of other products such as stainless-steel cooking-ware and glass lantern globes.

By June 1973, the company had incurred losses during each year of its operations except the most recent and the balance sheet for that fiscal year showed a total accumulated loss of Rls. 32 million, over one-half of its original equity capital. According to the Iranian commercial code, it had been legally bankrupt for over a year, but in the absence of any claims by creditors, it had continued operations. Exhibit 1 shows the company's balance sheets for all years of operations and Exhibit 2 shows the profit and loss statements for those years. Despite a forecast of a small net profit for the coming year, the company would still be in a state of impending bankruptcy by the end of fiscal 1974.

Company History. In 1965, the Tehran-Tokyo Trading Company was a subsidiary of the Behshahr Industrial Group and had been importing thermos flasks and other consumer products from Japan for a number of years. The thermos flasks, manufactured by Eagle of Japan, and marketed under the brand name "Eagle" were large-mouthed vacuum thermos bottles and were the Tehran-Tokyo Trading Company's most successful product. Employing a double walled glass vacuum bottle for insulation, these "ice-flasks" were sold in sizes ranging from 4 pints to 10 pints. They were widely used by consumers for storing ice in the absence of home refrigerators, keeping fruit and beverages cold, and for carrying ice and cold foods on family outings. The Eagle ice-flask was the only product of its kind on the market and the Eagle brand name became well known, as the product was common in many Iranian households.

Copywrite © 1974 by the Iran Center for Management Studies.

In 1966, a six pint flask cost Eagle of Japan about Rls. 200 to manufacture and sold in Iran for a retail price of about Rls. 480. The Tehran-Tokyo Trading Company's margin was about 30%. Mr. Mohammadi, the managing director of the Tehran-Tokyo Trading Company, reasoned that the company's profit margin would be substantially increased if the flasks were locally manufactured and, therefore, began negotiations with Eagle of Japan to establish an Iranian flask manufacturing venture.

In 1966, the Eagle Manufacturing Company was formed as a joint venture between the Tehran-Tokyo Trading Company and Eagle of Japan. Its initial equity capital was Rls. 15 million, 51% of which was supplied by the Tehran-Tokyo Trading Company and 49% by Eagle of Japan. In addition to this equity capital it was agreed that debt capital would be supplied by the Tehran-Tokyo Trading Company. The agreement also stipulated that Eagle of Japan would supply four Japanese experts for two years to train Iranian workers in the assembly of flasks.

Integrated Manufacturing. Mr. Mohammadi's original plan was to sub-contract the manufacture of the double walled glass bottles and their metal containers. He estimated that when production got under way, manufactruing costs would be Rls. 171 per completed flask. As soon as construction on the new plant began, Mr. Pourkamal cut the retail price on the imported flasks from Rls. 480 to Rls. 275 as a tactic to destroy competition and pre-empt the manufacturing strategy.

After testing the first samples of glass bottles received from sub-contractors, the Japanese experts reported that the local soda-lime glass was too soft and that all of them would break either in the vacuuming process or when exposed to hot or cold foods. They said that in place of the local glass, a hard (i.e. heat resistant) borosilicate glass, requiring a special high temperature furnace, was needed to produce the glass bottles. No Iranian contractor could be found to supply borosilicate glass, and Mr. Mohammadi decided that EMC would produce its own glass bottles. At the same time, Mr. Mohammadi discovered that no local metal shop was willing to supply the metal containers for less than Rls. 275 per unit and he decided to produce the container in-house as well.

To finance this increased manufacturing activity, equity capital was raised to Rls. 30 million. Eagle of Japan claimed that the laws of Japan would not allow them to increase their commitments to the venture and the entire increase in equity came from the Tehran-Tokyo Trading Company, reducing the Japanese share to 25%. The original agreement that Eagle of Japan receive 2% of manufacturing costs as royalty was retained but the technical assistance was withdrawn.

Crisis in Production and Profit. EMC had difficulty in finding anyone in Iran with enough knowledge of glass technology to melt borosilicate glass. They finally located and employed a White Russian glass-maker who had left the Soviet Union after the Revolution and hired him to construct the necessary furnace and to supervise the melting of the glass. It was found that the locally made refractory materials[1] melted at a slightly lower temperature than the glass; the glass became contaminated with the refractory materials and in about one month's time the furnace became unusable. A new furnace, costing

[1] Refractory materials are temperature resistant ceramics used to construct furnaces.

Rls. 40,000 had to be constructed each month. Because the glass was contaminated, breakage of the flasks was high during vacuuming and out of one thousand starts only 70 to 80 flasks went to assembly.

Mr. Mohammadi's original estimate had been that production would be at the level of 300,000 units per year. During the first six months of operations, the company managed to produce only 2000 flasks. When the Russian glass expert told Mr. Mohammadi that the solution to the technical problem was to buy a foreign furnace at a cost of Rls. 15 million, Mr. Mohammadi fired him.

By mid 1968, the Board of Directors became concerned with the company's performance and the way that Mr. Mohammadi had been handling its affairs. By this time, the company had incurred debts to the Tehran-Tokyo Trading Company of Rls. 90 million in addition to the Rls. 30 million in equity that had been invested. Mr. Mohammadi was reluctantly forced to sell his interest and leave the company.

Mr. G Ladjevardi, chairman of the Board of Directors, approached Mr. Fahimi, a 29 year-old M.B.A. graduate from the University of California at Berkeley, and asked him to take the position of Production Manager. Mr. Fahimi had previously been an assistant controller of Behshahr in charge of budgeting, feasability studies, capital budgets, and special reports. Mr. Fahimi accepted the position and a glass technology consultant was assigned to help him with the technical details. Mr. Mortazavee, the Deputy Director of the Tehran-Tokyo Trading Company and a relative of Mr. G. Ladjevardi, was placed in charge of the company's sales, finance, and administration.

Mr. G. Ladjevardi then raised the company's capital by changing some of its debts to the Tehran-Tokyo Trading Company to equity capital. Mr. G. Ladjevardi decided to order a German furnace and French refractories for Eagle at a cost of Rls. 15 million. The useful life of this furnace was estimated to be five years from the date of installation. Since it would take 18 months for delivery of the furnace, a temporary furnace was errected from used refractories for use during the waiting period. This furnace could only melt middle-hard glass from which flasks could be produced with a breakage rate of about 80%.

In the first quarter of 1969, the Ministry of Finance called for a tax audit and rejected the company's books because of minor erasures of accounting errors. Although the company had been operating at a loss, it was asked to pay taxes. Management used this opportunity to set up a cost accounting system. As a result, it was discovered that the company's unit cost was more than Rls. 275. Also, the books showed negative retained earnings of Rls. 40 million and, according to the Iranian Commercial Code at that time, if a company's accumulated loss exceeded one-half of its original equity, it was legally bankrupt in the event of any claims by creditors. To avoid having to close its doors, accounting changes were made, the value of inventory increased, broken glass was not written off as a loss, and the furnaces which became useless after a month of use were depreciated over a number of years. In this manner the company was able to reduce its accumulated loss to Rls. 10 million.

The "homemade" furnace was completed in April 1969 and produced 80,000 flasks before it was replaced by the imported furnace in the summer of 1970. In the six months that followed, 140,000 flasks were produced. The breakage rate during the period was about 65%.

In 1969, the Eagle Manufacturing Company began to produce Coleman thermoses under a license agreement with the Coleman Company, U.S.A. The Coleman jugs were large, wide-mouthed containers that used a plastic compound as an insulating material. While the temperature insulation a Coleman thermos provided was less effective than that of a double glass vacuum bottle, their low cost, larger capacity and greater sturdiness made tham popular with consumers.

In the Spring of 1970, Mr. Fahimi decided to utilize the excess capacity of the glass furnace to produce "tea-flasks" at the rate of 2000 per month. These tea flasks were much smaller than the Eagle ice flasks, having a capacity of about one quart. While they were too small to be used as food containers, tea flasks had found a ready market for use on picnics and among workers and farmers who wished to have hot tea available during their work breaks. Shortly after introducing the tea-flasks, Mr. Fahimi decided to utilize the excess capacity in the company's metal shop by taking outside orders on metal products, such as stainless-steel pots and pans.

Crisis in Sales. Until 1971, the company's ice flasks were sold to wholesalers almost as fast as they came off the production line; management saw no reason to concern itself about the market for this product. A cost analysis had showed that the break-even point was about 120,000 units per year, and the market was estimated to be about 300,000 units per year.

In September 1971, Mr. Fahimi's duties were expanded to include finance and sales. Alarmed at the large inventory of ice flasks in the company's warehouse, Mr. Fahimi decided to survey the market. A questionnaire was sent to 2000 retail outlets carrying the flasks. After analyzing the response to this questionnaire, Mr. Fahimi realized that the market in Khuzestan, up to then the company's largest, had all but disappeared and sales were also down in most of the other provinces. The survey showed that in addition to the company's Rls. 40 million inventory of finished flasks, an inventory of Rls. 15 million had piled up on the retailers shelves. The steadily increasing use of refrigeration by Iranian households, Mr. Fahimi reasoned, was eroding the market for ice flasks, although neither the manufacturer nor the retailers had been aware of this trend.

Mr. Fahimi went to the Board of Directors of EMC and convinced them that the company should quit the production of ice flasks and concentrate on tea flasks and Coleman brand thermoses. Mr. Fahimi also suggested that the price of the ice flasks remaining in inventory be raised by 50% because consumers did not purchase these flasks often enough to keep track of the price and were relatively insensitive to price differentials within a range of 30 to 40 tomans. The board agreed; production of Eagle ice flasks was stopped, the price was raised by 50% and a luxury table flask was introduced to supplement the flask line.

Sales, Marketing and Competition

Both the "Eagle" and "Coleman" brand names enjoyed excellent acceptance by consumers in Iran. Because these products had been imported and distributed long before local manufacturing began, their brand names had almost become generic terms for the products they represented. Until mid fiscal 1973, EMC had sold all of its products through wholesalers (bonakdar) and did very little advertising. Thus, customer familiarity with these products was due mainly to word-of-mouth "advertising" and direct retailer promotion.

Inorder to simplify its marketing and relieve the burden of carrying a large inventory of finished goods, EMC reached an agreement toward the end of fiscal 1973 with an affiliated company, the Arien Trading Company, for the distribution of many of its products. According to the terms of this agreement the entire output of flasks was purchased by Arien and sold through their sales and distribution organization. Arien did not mark-up the flasks it sold to distributors. Mr. Fahimi believed that this arrangement would have to be retained for as long as EMC was unable to finance its inventory of finished goods. Exhibit 4 shows the channels through which tea flasks moved from Eagle to the consumer and the approximate margins received at each stage.

Because of its methods of distribution, the company's sales figures did not accurately reflect consumer response to EMC's products, but rather the purchases of Arien Trading Company and other wholesalers. Wholesalers' purchases, in turn, did not always indicate high demand. Eagle and Arien offered seven month credit terms to wholesalers and retailers who bought from them. Occasionally, Mr.Fahimi discovered Eagle flasks retailing at prices that were lower than the factory price. He explained:

> Sometimes, wholesalers use our flasks as a source of financing. They buy them from us on credit, sell them immediately at a discount for cash, and have six or seven months to use the money before they have to pay us. When this happens a flask we sold for 126 rials might appear on a retailer's shelf priced at 125 rials! While I don't think that this kind of behavior is really rational, it does happen, and its difficult for us to ascertain whether demand is really for our products or for our financing.

Sales of tea flasks and Coleman type thermoses together accounted for over 90% of the company's revenues in discal 1973. Exhibit 5 shows sales of all products for that year.

While EMC did not distribute its own products, it did incur sales costs due to the discounts offered to the wholesalers. Sales costs during fiscal 1973 amounted to Rls. 5.8 million or approximately 6% of total sales for that year.

Ice Flasks. By the end of fiscal 1973, sales of Eagle ice flasks had dropped to 5, 818 units (Rls. 2.4 million) or less than 3% of total company revenues. Management estimated that ice flask sales would continue to decline gradually. Management presently viewed the decline in ice flasks

as due to the increased use of home refrigerators and had no plans to try and revive sales. Production of ice flasks had stopped in 1971 and the current inventory was expected to last about another 2 years.

Tea Flasks. Sales of tea flasks had increased rapidly since the product's introduction in 1969. Sales had increased from 360,000 units (Rls. 45 million) in 1972 to over 470,000 units (Rls. 60 million) in 1973. Management forecast slightly higher unit sales for fiscal 1974, based on the assumption that no major changes in the competitive environment or market conditions would occur.

While EMC was the only local producer of tea flasks, similar products were imported from a number of countries. In all, management estimated that over 700,000 tea flasks were sold in Iran during their fiscal year of 1973, of which 60% were EMC's. As can be seen from Table 1, almost all imported tea flasks had CIF prices that were lower than EMC's average ex-factory price of Rls. 126 per unit. Table 2 shows how customs duties and commercial taxes increased the cost to the wholesalers of imported flasks. Even after these duties and taxes, the tea flasks imported from the Republic of China cost the wholesaler about Rls. 108 per unit. Mr. Fahimi maintained that it was competition from the Chinese flasks that kept EMC from being profitable:

> Even though we have 60% of the market, low priced Chinese flasks "dumped" in Iran prevent us from pricing at a level that would return a decent profit. If it weren't for competition from this source, I am sure we could raise our prices by 20 to 40 rials without cutting the volume.

Both the Chinese flasks and EMC's flasks retailed for about Rls. 152, so that wholesalers and retailers received about Rls. 32 (30% mark-up) for each Chinese flask they moved and only Rls. 26 (21% mark-up) for each Eagle flask. However, the importation of flasks usually required cash payment and making an import deposit with the Central Bank while EMC provided comparatively easy credit terms.

Management's first reaction to the Chinese' dumping had been to go to the Ministry of Economy and apply for protection. The Ministry had responded by changing the tariff schedules but the effective change in the costs of importing the flasks did not give EMC the competitive advantage they had sought. According to the new tariff schedules for 1973, the customs duty was changed from Rls. 40/kg. to 40% of value and the commercial tax was raised from Rls. 60/kg. to Rls. 72/kg. These changes raised the total tariff rates on tea flasks from China by Rls. 5.8/unit.

British flasks were higher priced than EMC's. Consumers generally considered the British flasks to be superior to Eagle flasks and they were purchased by people who were more concerned with quality than price. Although EMC's flasks had been patterned on the British type and had the same general appearance, the British products were able to keep a significant market share because of the consumer preference for quality imported goods.

A considerable institutional market also existed for tea flasks. Fancier offices, hotels and restaurants tended to buy the more expensive British and U.S. flasks while there was a market for more reasonably priced flasks

among the less image conscious establishments and hospitals.

In explaining why EMC's market share remained substantial in the face of Chinese dumping and European imports of higher perceived quality, management stressed four factors: (1) Importing flasks required cash payments and making an import deposit with the Central Bank while EMC gave liberal credit terms; (2) Imported products were not always available and the lead time between order and receipt could be quite long; (3) EMC's flasks had a better appearance than the Chinese imports and were fully comparable to the high-priced European flasks; and (4) Chinese flasks tended to be purchased for indoor use while consumers preferred the more attractive Eagle flask for outdoor use.

Coleman Thermos. During fiscal 1973, EMC sold nearly 100,000 Coleman thermoses at an average unit price of Rls. 290. This volume was estimated to be 30% of the local market and Coleman sales accounted for 29.2% of the company's total revenues.

Coleman thermoses were originally imported into Iran by the Arien Trading Company which presently acted as EMC's distributor. With the advent of local production, importation of Coleman type thermoses stopped and there were about 20 local producers making Coleman type thermoses in 1973. EMC was the largest producer and the next largest had a volume about 70% of EMC's. The remaining 19 producers had very small individual market shares.

The proliferation of producers of Coleman type thermoses in Iran was due largely to the relatively simple technology, the low capital requirements, and high local demand. EMC's management felt that it was the leader in the production of thermoses in terms of price, quality, brand image and market share. They claimed that consumers tended to think of their competitors' products as "cheaper copies" of the Coleman thermos.

EMC was the only Iranian company permitted to use the Coleman brand name. While management recognized that consumers viewed the Coleman name as representing the highest quality, the product standards imposed by the Coleman company made EMC the high cost producer. Expensive special insulating materials had forced EMC to price its thermos at Rls. 290 while the average ex-factory price of competing products was Rls. 230-270.

Coleman thermoses were distributed in much the same manner as EMC's tea flasks. The channels of distribution are shown in Exhibit 6. Management had no information on the regional distribution of Coleman sales and was not able to estimate the breakdown between rural and urban sales. Mr. Fahimi was certain, however, that Coleman sales would grow at least as rapidly as national income.

In addition to the local market, there was a possibility that EMC would be able to open up a significant export market for Coleman thermoses in the Persian Gulf states and other neighboring countries. There were, however, a few difficulties involved in opening up these markets. Some of the Gulf states banned the import of plastic products from Iran because a number of Iranian plastics producers were of the Jewish faith and such purchases were considered "helping the enemy". Turkey and Pakistan usually did not import goods from Iran except under barter agreements. Hard cash purchases by these

countries were possible only at very low prices. Mr. Fahimi believed that since EMC could offer thermoses at a price considerably lower than that of U.S. imports, these barriers were not insurmountable.

Due largely to transportation costs, the CIF price for U.S. made Coleman thermoses was considerably higher than the $4.45 export price that EMC would be able to offer. In addition, for each unit exported, EMC could receive Rls. 6.3 from the government as a reimbursement of duties charged on imported components and materials.

EMC had had numerous solid opportunities to sell Coleman thermoses in the gulf area and had recently received an order for 40,000 units. There were several factors, however, which management had to weigh in trying to decide whether to pursue this export market. The licensing agreement with Coleman U.S.A. restricted them to the Iranian market. Coleman U.S.A. was presently exporting to the Persian Gulf States and probably would be reluctant to allow EMC to sell in that region. If EMC were to break its licensing agreement with Coleman U.S.A., there was a strong likelihood that they would not be allowed to continue to use their brand name (despite the fact that their products were labled with the spelling COLMAN instead of the correct trade name spelling COLEMAN, due to an error when they first started production). Mr. Fahimi believed that marketing the thermos under a different brand name would reduce its appeal.

A second argument against severing ties with Coleman U.S.A. was the possibility that the company might want to expand its line. Coleman U.S.A. had expressed interest in cooperating with EMC in the production of picnic and sports products. The market potential in Iran for such products appeared excellent. The only obstacle was that these products were not related to EMC's present output and the government might not agree to grant the necessary permission to expand.

Finally, EMC was operating at near full capacity at its present rate of 100,000 Coleman thermoses per year. Given the restrictions on expanding its manufacturing facilities (see page 12) EMC would be faced with the need to divert domestic sales if it decided to pursue the policy of building an export market.

Production

EMC's plant was located north-east of Tehran, about 20 meters outside the city limits. Exhibit 7 shows the plant layout and Exhibit 8 gives the size of the workforce in each production section. EMC was the only manufacturer in Iran with the capability of producing the hard borosilicate glass used in the company's flasks and other glass products. In 1973, its furnace was operating two shifts per day and produced 800 kg. of molten glass per shift.

The forming operation for flasks took several steps. First the outer wall of the flask was blown by hand. It was them mounted on a semi-automatic machine where the inner wall was formed and sealed. A tube, later to be used in the silvering and vacuuming processes, was then connected to the flask at the bottom.

At the vacuuming station, a chemical solution containing silver was poured into the cavity between the two glass surfaces. The glass was then rotated on a special machine, insuring that the solution was evenly distributed over the inner surface of the glass. The remaining solution was then extracted, leaving a fine film of silver, giving the flask better insulating qualities (a silver surface reflects heat). The flasks were then moved to the vacuuming machines. Pumps reduced the pressure within the cavity to about 0.1 atmospheres and the cavity was sealed. In the next step, boiling water was poured into the thermos and the neck was blocked with a cork. Twenty-four hours later, a worker tested the temperature of the water in each flask by uncorking it and letting the water run over his fingers as he emptied it. If the temperature had dropped below 45°C the part was rejected. The accepted parts were then moved to the assembly section where the flask was placed inside a protective metal container and the cork and cap added to form the finished product.

During the early years of operations, excessive breakage in the glass forming and vacuuming operations had resulted in very high production costs. Following the installation of the two semi-automatic glass forming machines in the glass section in early 1973, the breakage rate was reduced from nearly 60% to about 25%. Exhibit 9 shows the breakage rate at the various stages of production of ice flasks for late fiscal 1973. Exhibit 10 gives a manufacturing cost breakdown for a tea flask for the same period.

According to Engineer Bahreini, Production Manager, none of EMC's glass products were produced at a high enough volume to take advantage of the economies of scale that the larger international producers of these products enjoyed. Almost all the competing foreign flasks imported into Iran were produced in fully automated factories. Mr. Bahreini explained that hand forming of flasks was not only more expensive, but also resulted in an inferior product. Approximately 3% of EMC's flasks were returned to retailers because they did not insulate properly. While EMC was willing to re-imburse wholesalers and retailers for returned merchandise, management felt that the annoyance factor due to customer complaints made retailers less willing to carry Eagle flasks. The Ministry of Economy had offered full import protection for EMC's flasks if the company would automate fully and raise production to three million units per year, exporting whatever it could not sell within Iran. It was estimated that this would require an investment of approximately Rls. 70 million. In light of present local demand and the low price of competing Chinese flasks, Mr. G. Ladjevardi was convinced that EMC would not be able to sell an output of three million units.

Coleman Thermos. Coleman thermoses were assembled in a shop on the second story of the factory. The Coleman thermos was a totally non-glass product. The insulating characterisrics of this product were achieved by forming a layer of polyurethene foam between its inner and outer walls. The chemicals used in producing this foam were imported in liquid form. A carefully measured amount of the mixture was placed between the two plastic walls of the thermos and then the unit was moved into a heating oven. The amount of heat needed to change the liquid to a foam was very critical and required accurate regulation.

Engineer Bahreini explained that EMC was the only local producer using polyurethene in its thermoses. All other producers used a less expensive foam with inferior insulating qualities. In June 1973, it cost EMC Rls. 279 to produce a Coleman thermos. Engineer Bahreini estimated that competitors' costs

were approximately 20% less than EMC's Exhibit 11 gives the manufacturing cost breakdown for Coleman thermoses.

The Problem of Over-Capacity. EMC's factory and equipment were operating well under capacity. Production of glass could be tripled by moving to a three shift operation and doubling the number of forming machines. To increase the number of forming machines, alterations would have to be made in the plant building, and supervisory staff in the glass making section would have to be increased if the company decided to go to three shifts per day. Considerable savings in fuel and overhead could be made, however, because the furnace temperature had to be maintained even during idle periods. Lack of strong market demand for the company's present glass products had thus far kept EMC's management from seriously investigating capacity expansion plans.

In addition to over-capacity in the glass making section, EMC's metal tool, paint, and assembly section were operating one shift per day. In response to this over capacity, management had tried to diversify into other products. Mr. Fahimi reasoned that in addition to the profits that could be realized from new products, the costs of producing the present line of products could be reduced substantially by spreading overhead over a greater volume of product

One product that had been added as a result of this strategy was stainless steel cookware. EMC's line of cookware consisted of two sizes of frying pans and three sizes of stewing pots and went into production in 1971. The body of these pots was formed from sheet stainless steel on a 200 ton hydraulic press which had originally been acquired to fabricate the larger metal portions of the Coleman thermoses. The results of this diversification did not reach expectations. Mr. Fahimi believed that while EMC's cookware retailed for approximately 40% less than imported cookware of the same type, the market for these products in Iran was undeveloped. The Iranian consumer usually bought aluminium cookware and was unaware of the superior quality of stainless steel. Aluminium was about half as expensive as stainless steel kitchenware. The fact that stainless steel is better to cook on (superior heat conductivity than aluminium) and that its useful life is much longer was not known by the typical housewife.

According to Engineer Bahreini, imported cookware of the type produced by EMC was of higher quality. The problem, he explained, lay in EMC's method of production. The pressing operation exerted strong stresses on the bottom portion of the pot being formed. This caused the metal to be stretched thin, reducing the superior heat transfer qualities of the stainless steel. A superior method of production, he explained, would be a "spin process". He estimated that it would cost approximately Rls. 4 million to purchase the necessary equipment to acquire this capability.

Iran imported numerous "hard glass" products, such as glass laboratory equipment, "Pyrex" type glass cookware, non-breakable drinking glasses, and lantern glasses. EMC had already decided to go into the production of "Zanbouri"[1] lantern glasses. Sales of Zanbouri glasses in fiscal '73 had amounted to 11,500 units. Management planned to begin the manufacture of a larger size Zanbouri glass during fiscal '74.

---

[1]The term Zanbouri designated any of a variety of lanterns which operated by forcing pressurized kerosene vapor over a woven silk wick. The heat from the combustion of the kerosene caused the wick to glow with an incadescent white light.

EMC's management had also considered going into the production of glass laboratory equipment. The total market for such glassware in Iran was estimated to be around Rls. 150 million and was expanding rapidly. Most manufacturers of glass laboratory equipment produced a full line of such products. EMC's management believed that the market for the more common type of laboratory equipment would be large enough to warrant the investment for the necessary shaping equipment. However, they were reluctant to enter this market because they also believed that the purchasers of such equipment preferred to stay with one brand name for all their needs. This factor would force them to become a full line producer and the lower volume of the less frequently purchased items would not allow them to be competitive with the larger international producers.

Other Possible Products. EMC had considered a number of other products which could be produced with their present facilities or without major new capital investments. No decision had been made on most of these products either because of lack of adequate market information or other limitations.

One possibility was diversification into Coleman brand sports and picnic equipment. Coleman, U.S.A. had expressed interest in cooperating in such a venture. A second possibility which had been considered was the production of loose fiber glass. Fiber glass could be produced as a by-product of their present line of glass products. The market in Iran was estimated at 3,000 kg. per day. The current price was Rls. 50/kg., while production costs were estimated at approximately Rls. 15/kg. Management roughly estimated that it would require an investment of Rls. 50 million to acquire this capability.

A third possibility was the production of vitrocolour glass, a small square of colored glass used extensively for construction in Europe and the United States. Like fiber glass, it could be produced with EMC's present furnace and would require no significant additional investment. Unfortunately, vitro color glass had been imported into Iran only in token amounts and management had no way of accurately assessing its market potential.

The company had also decided to integrate backwards into plastics. The plastic parts for its products were presently being sub-contracted. Management estimated that the investment required would be Rls. 10 million at most, while annual savings would approach Rls. 6 million. The technology required would not be beyond the company, and it already owned the necessary dies (currently being used by the subcontractors.) It was estimated that by integrating into the production of plastics, EMC would save Rls. 32 per Coleman thermos and Rls. 5 per tea flask.

Limits to Diversification. One factor which affected any plans that Mr. Fahimi might make to diversify was the location of the factory. In 1966, when Mr. Mohammadi and Mr. G. Ladjevardi had decided on the site for their new factory on the outskirts of the city, the surrounding area had been largely undeveloped. In the intervening years, Tehran had grown considerably and the environs of the factory had become primarily a residential district. The market value of the plant site, which had been purchased for Rls. 2.1 million, was estimated to be at least 20 times that figure in 1973.

By 1973, the government was encouraging businesses to move their factories away from the immediate vicinity of Tehran. Mr. Ladjevardi believed that government permission would be withheld for any diversification into products that were not part of EMC's current lines or which would require the importation of capital equipment. According to Mr. Fahimi:

> We can diversify into other products as long as we don't make any major changes in the nature of our business. As I understand it, adding products such as Coleman brand sports equipment may not be permitted, both because we would have to import the machinery necessary to produce these products and because sport equipment would be a big departure from the type of goods we are already producing. The alternative would be to re-locate our factory in some provincial city. I don't think anyone has seriously considered this alternative and I doubt Mr. Ladjevardi would be willing to take such a step if it meant making further investments in the company.

Tightening the Screws. Concurrent with the search for alternate products, Mr. Fahimi attempted to lower costs through reducing overhead and the size of EMC's labor force. Mr. Fahimi saw this effort as essential to holding down the prices of EMC's products in the face of rising labor and raw material costs. According to company records, EMC's labor force of 275 workers received Rls. 1.79 million in wages and produced 36,000 flasks and 10,500 Coleman thermoses in June 1972. In June 1973, 259 workers received Rls. 1.73 million and produced 64,000 tea flasks and 17, 500 Coleman thermoses. Mr. Fahimi predicted that by June 1974, the labor force would be down to 200. This reduction in labor force had been achieved through normal attrition and no workers had been forced to leave. Improvements in production methods had allowed EMC to keep production rates up. Mr. Fahimi's efforts to reduce administrative, distribution, and financial expenses had not been as successful. These expenses amounted to Rls. 14.3 in fiscal '72 and rials 15.4 million in fiscal '73.

Mr. Fahimi's Decision. In summing up his position, Mr. Fahimi stated that he saw several alternatives open to EMC. One was to increase capacity utilization through the addition of selected products. A more radical solution, but possibly a viable one, he explained, would be to withdraw completely from the production of glass products and concentrate on Coleman products. He explained that this would most likely entail the sale of the present factory and a move to a much smaller location.

As of June 1973, Mr. Fahimi was still on the verge of leaving the company. As he explained:

> The company showed a modest profit this year. It can probably continue to make a modest profit and perhaps improve somewhat. It must be remembered, however, that EMC's profit and loss statement overstates the company's real profitability. The loans we have received from the Behshahr Group carry very low interest rates and are a net loss to the Group. The cost of carrying our finished goods inventory is borne by Arien Trading Company and Arien does not earn any profit by handling our products. At this rate it is doubtful if the owners will ever recover their investment much less earn a return.

I would like to turn the company around. In the six years since we started operations, we have learned a lot of hard lessons. EMC has acquired a lot of strengths both in production and in the market place. Our products have acquired a good name with consumers. We have the capability of going into any number of products that have good potential in Iran.

The problem is due to other factors; the range of options open to me is limited. Mr. G. Ladjevardi is reluctant to put any more money in until the company has proven itself, and I can't say that I blame him. I am sure that he would invest if I could show him a sure way of making EMC profitable, but given the fact that we are operating in such a competitive environment I can't even convince myself of that.

I like the company, I like my bosses and I would like to stay. I'm certain that I can stay, but frankly, given all the limitations imposed by EMC's spotty history and the nature of the environment I'm not sure but what I would be doing myself a disservice if I did.

Exhibit 1

EAGLE MANUFACTURING COMPANY

Summary of Balance Sheet, 1967-1973

(Rials in millions)

ASSETS	1967	1968	1970[1]	1971	1972	1973
Cash	0.1	0.1	2.7	0.2	1.0	0.4
Receivables	6.9	8.2	0.3	5.5	4.6	2.6
Trade debtors	5.0	0.8	6.5	6.9	8.0	8.3
Deposits	2.9	5.7	9.7	4.8	3.4	2.7
Inventory						
Raw Material	8.8	24.0	28.2	41.6	31.5	28.7
WIP	9.5	10.9	8.4	15.5	14.7	15.6
FG	3.4	11.7	27.0	28.8	0.9	0.5
Total Current Assets	36.6	61.4	82.8	103.3	64.1	58.8
Fixed Assets	41.8	52.0	72.5	76.7	82.6	83.5
Less Depreciation	1.2	7.5	15.0	18.7[2]	18.7[2]	29.2
Net Fixed Assets	40.6	44.5	57.5	58.0	63.9	54.3
Total Assets	77.2	105.9	140.3	161.3	128.0	113.1
LIABILITIES						
Notes Payable	9.2	4.0	5.1	7.5	7.7	10.0
Overdrafts	5.4	7.5	35.7	43.8	45.4	65.3
Creditors	21.3	31.1	49.3	72.6	38.5	5.7
Advances	1.0	4.1	2.2	2.4	8.8	4.1
Bad Debts	-	-	0.5	0.5	0.5	-
Total Current	36.9	46.7	92.8	126.8	100.9	85.1
Long Term Debt	20.0	20.0	12.2	4.2	-	-
Capital	30.0	60.0	60.0	60.0	60.0	60.0
Retained Earning	(9.7)	(20.8)	(24.7)	(29.7)	(32.9)	(32.0)
Total Liabilities and Equity	77.2	105.9	140.3	161.3	128.0	113.1

(1) In 1348, the company changed its fiscal year from Favardin-Esfand (March 21 to March 20) to Shahrivar-Mordad (Aug. 23 to Aug. 22). To effect this change, the fiscal 1970 included a 16 month period-- March 21, 1969 to August 22, 1970.

(2) To prevent accumulated losses from reaching half the capital, the company did make accounting changes that reduced depreciation charges by about Rls. 8 million in each of these years.

Source: Company Records.

Exhibit 2

EAGLE MANUFACTURING COMPANY

Summary of Profit and Loss Statements 1967 - 1973
(Millions of Rials)

	1967	1968	1970	1971	1972	1973
Sales	14.1	27.7	59.2	65.0	101.3	96.2
Cost of Goods Sold	13.5	32.2	50.6	58.1	89.1	79.0
Gross Profit	0.6	(4.5)	8.6	6.9	12.2	17.2
Administration	4.7	2.6	5.0	3.9	5.9	7.0
Selling	5.6	1.4	4.8	4.4	5.2	5.8
Interest	-	4.0	3.3	4.2	4.8	2.6
Other Income	-	1.4	(1.6)*	0.6	1.2	(1.0)**
Net Profit (Loss)	(9.7)	(11.1)	(6.1)	(5.0)	(2.5)	0.8

* Including is0.5 million rials due to writing off the last furnace

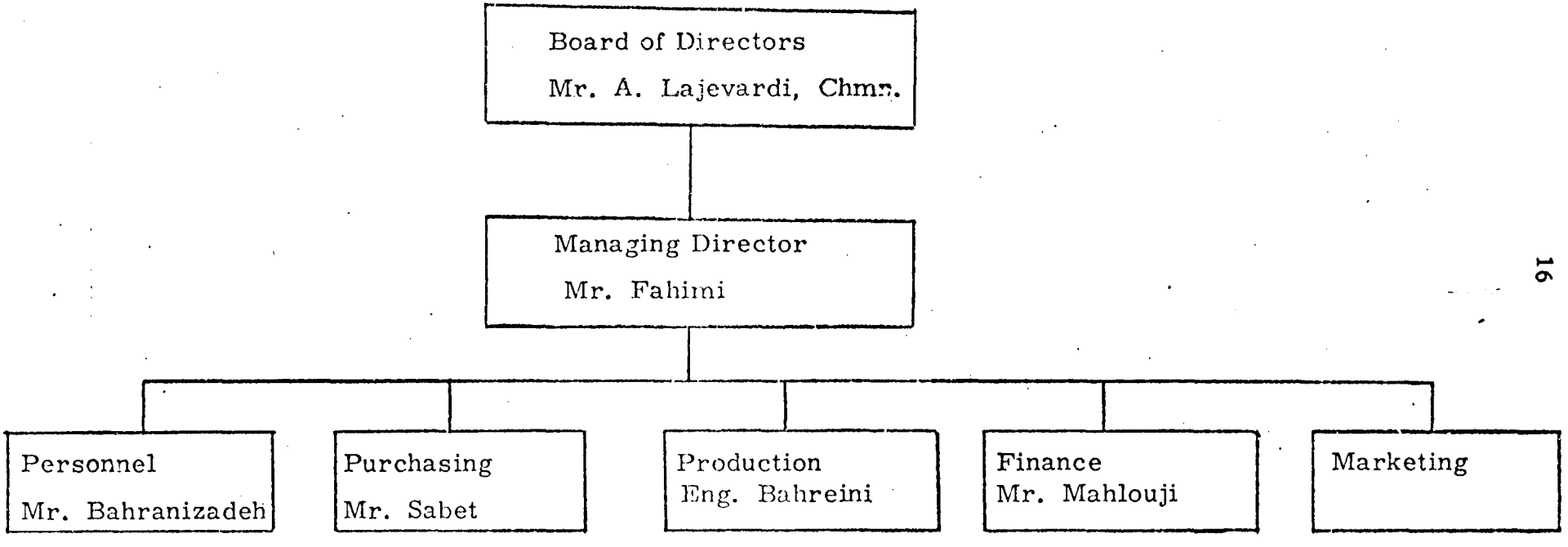
** Eagle sold part of its inventory of broken glass at below its book value of Rls. 5.85/Kg.

Source: Company Records

Exhibit 3

EAGLE MANUFACTURING COMPANY

Partial Organization Chart, June 1973



Source: Company Records

Exhibit 4

EAGLE MANUFACTURING COMPANY

Distribution Channels for Tea Flasks and Average Prices at Each Stage

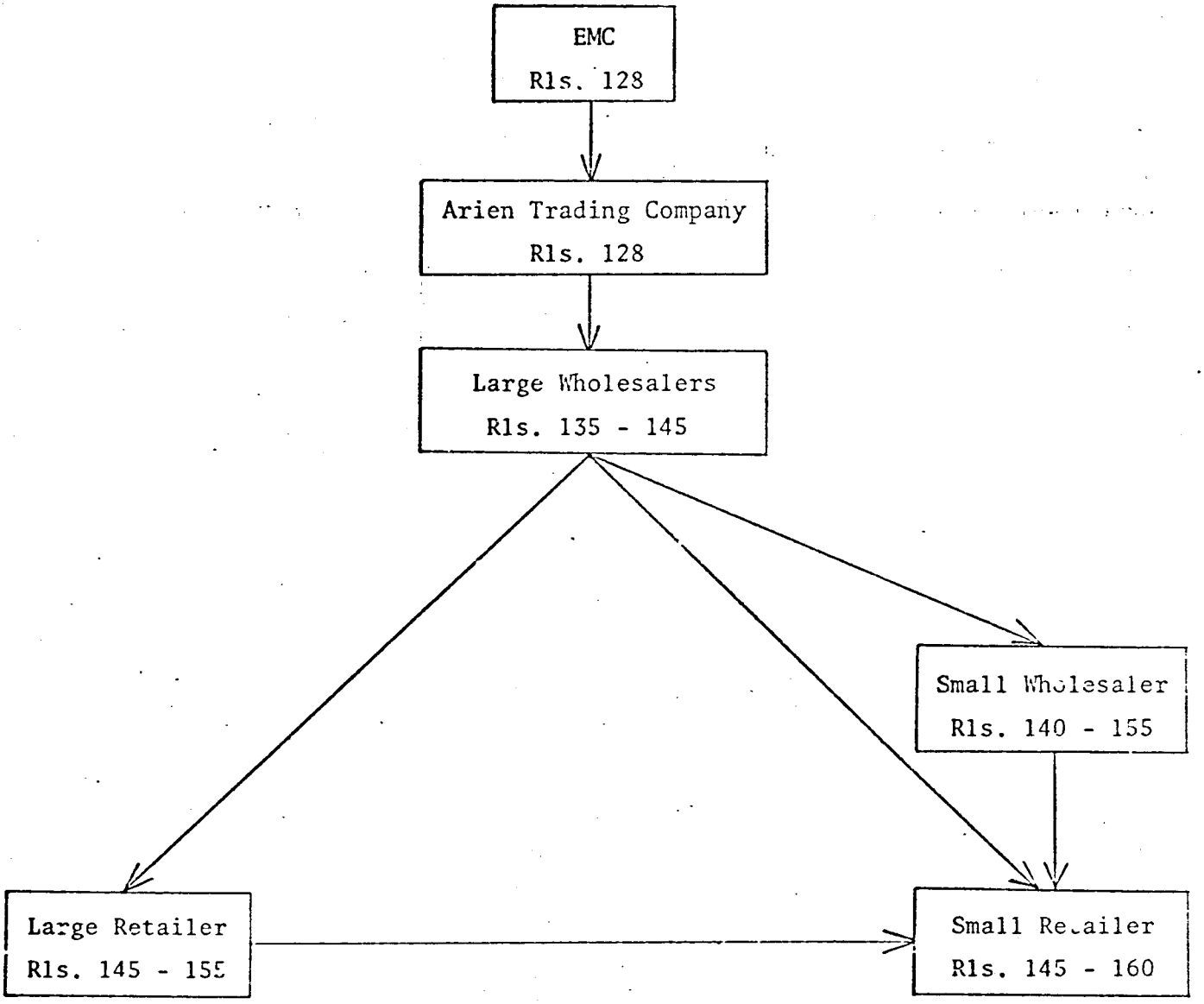


Exhibit 5

EAGLE MANUFACTURING COMPANY

Sales by product, Fiscal 1973 (rials)

Product	Units	Unit Cost	Unit Price	Total Sales
Tea Flask	347,242	90.26	127.67	44,331,786
Pitcher Tea Flask	115,480	122.18	121.37	14,015,323
Steel Tea Flask	7,907	231.06	252.49	1,996,430
Coleman Thermos	96,879	254.58	289.93	28,087,699
Ice Flask 300	225	316.72	241.15	54,259
Ice Flask 500	1,472	411.23	423.51	623,411
Ice Flask 600	1,134	488.01	549.96	623,653
Ice Flask 800	614	611.33	716.35	439,837
Reed Flask	2,373	290.28	289.35	686,629
Glass 2006	1,781	63.11	87.19	155,290
Glass 3006	1	115.25	160.00	160
Glass 5006	6,280	198.45	182.61	1,146,820
Glass 6006	2,822	238.88	270.29	762,750
Glass 8006	1,485	316.87	360.24	534,960
Zanbouri Glass, Sm.	11,493	13.11	37.17	427,195
Zanbouri Glass, Lg.	-	-	-	-
Pitcher Glass	337	65.56	90.00	33,930
Small Stewpot	2,162	296.21	349.45	755,510
Medium Stewpot	840	416.57	461.19	387,400
Large Stewpot	867	526.10	571.90	495,840
Small Frying Pan	755	254.62	347.97	262,710
Large Frying Pan	737	614.67	576.45	424,840
TOTAL SALES				96,246,432

SUMMARY:

	Sales	Cost of Goods Sold	Contribution
Tea Flasks	60,343,539	47,278,401	13,065,138
Coleman Thermos	28,087,699	24,663,456	3,424,243
Ice Flasks and replacement glass	5,027,769	4,797,638	230,131
Zanbouri globes and pitchers	461,125	175,389	285,736
Cookware	2,326,300	2,091,703	234,597
TOTAL	96,246,432	79,006,587	17,239,845

Source: Company records

Exhibit 6

EAGLE MANUFACTURING COMPANY

Distribution Channels for Coleman Thermoses and Average Prices at Each Stage

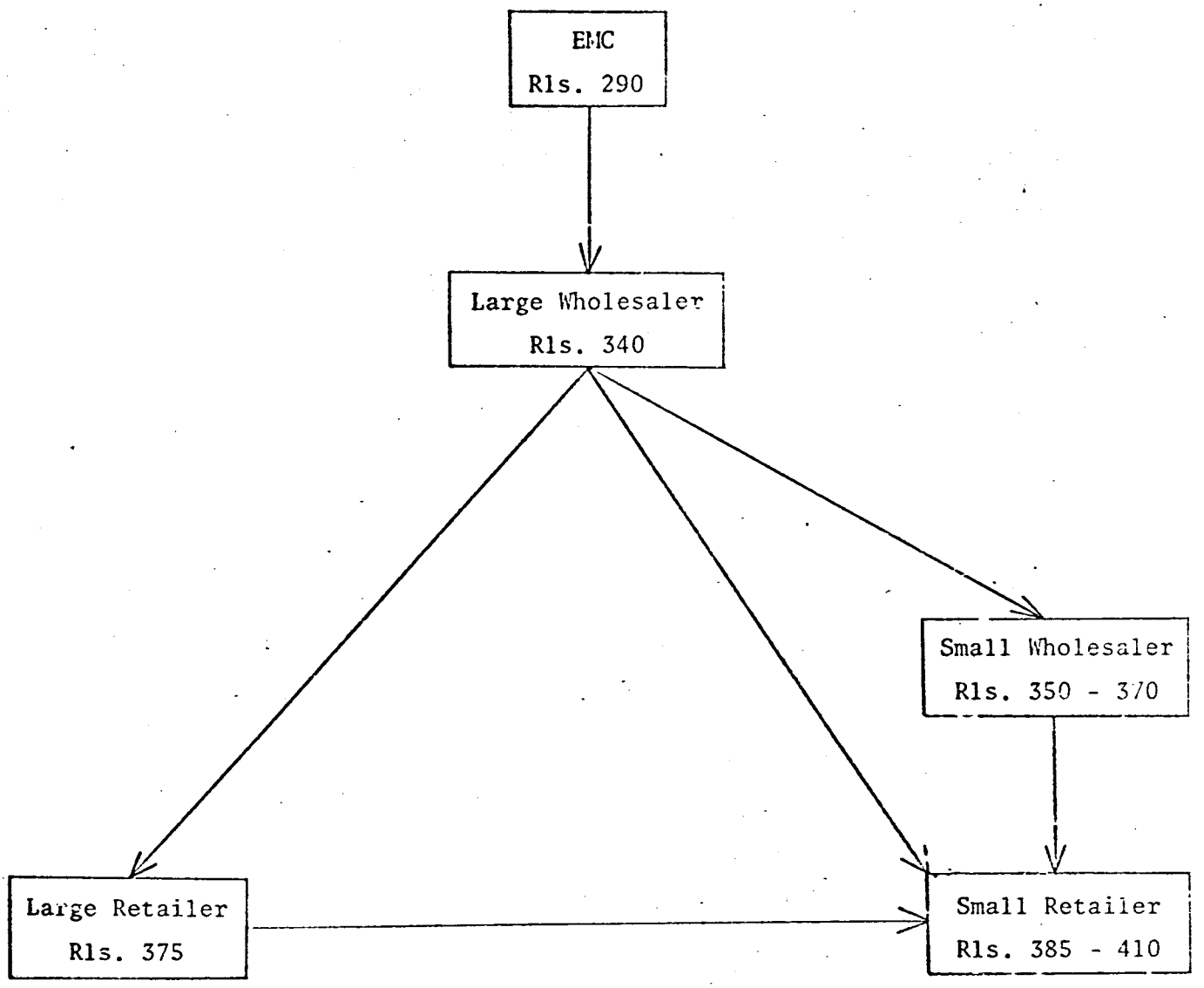


Exhibit 7

EAGLE MANUFACTURING COMPANY

Plant Layout

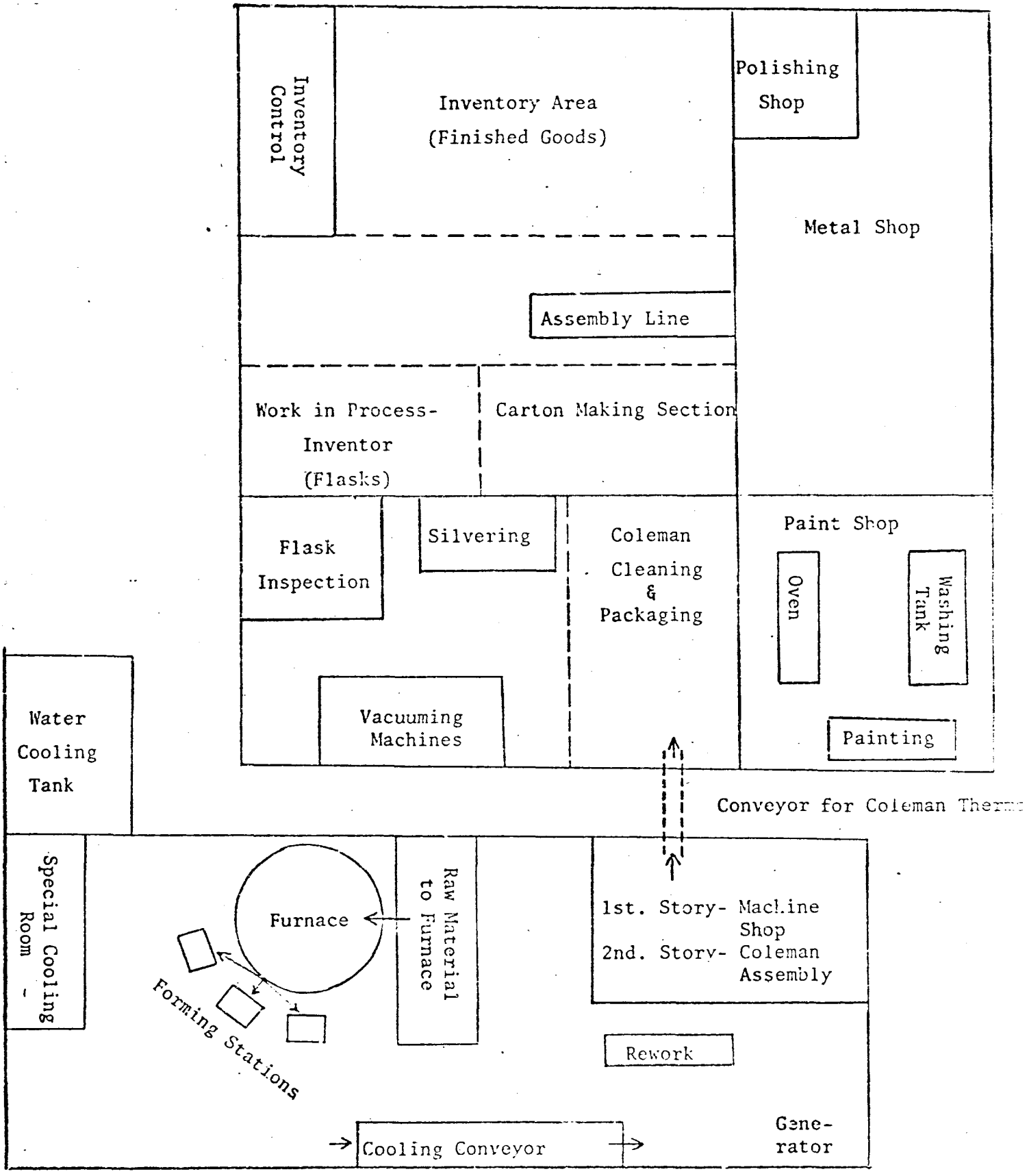


Exhibit 8

EAGLE MANUFACTURING COMPANY

Work Force in Each Section of the Eagle Plant, end of June 1973

Department	Type "A"	Type "B"	Type "C"	Type "D"	Total
1. Glass forming	39	38	11	2	90
2. Machine shop (tools)	-	-	3	1	4
3. Metal shop	4	5	20	1	30
4. Assembly, tea flask	-	-	25	-	25
5. Vacuuming	-	-	35	1	36
6. Paint shop	-	-	6	1	7
7. Coleman assembly	23	-	-	1	24
8. Warehouse	-	-	4	-	4
9. Polishing	-	-	4	-	4
10. Service	4	5	14	-	23
Total labor force					247
Administration					9
Production Staff					4
Total Employees					260

Type "A" - Morning shift, 6 AM to 2 PM

Type "B" - Afternoon shift, 2 PM to 10 PM

Type "C" - Day shift, 8 AM to 5 PM

Type "D" - Foreman

Source: Company Records

Exhibit 9

EAGLE MANUFACTURING COMPANY

Daily Glass Production, June 1973[1]

Broken Glass	1656.7 kg.
Arsenic	3.3
Potassium Nitrate	10.0
Total input to furnace	1670.0
Reduction due to melting process	5.0
Furnace output	1665.0
Spillage and long-pipe wastage	331.0
Glass to forming section	1334.0
Wastage in forming (10%)	134
Total, formed flasks	1200.0 kg.

Number of flasks @ 300 gm. each[2]	4000
Breakage in silvering and vacuuming (21%)	840
Number of finished flasks	3160

1) Glass was produced in batches, the furnace delivering approximately one metric ton of molten glass per batch. Undrawn glass had to be left in the furnace between shifts and kept at a temperature of 1200°C.

Prior to March 1973, the company had used a mixture of broken glass and raw materials to produce its glass. After that date, it began using exclusively broken glass with arsenic and potassium nitrate to retain the proper glass characteristics. In June 1973, EMC purchased arsenic at Rls. 28.21 kg. and potassium nitrate at Rls. 17.48 kg. The broken glass was inventoried at a value of Rls. 5.85 kg. At that time, Engineer Bahreini estimated that the company's inventory of of broken glass would last one year at the present volume of production.

2) Based on the production of small sized tea flasks.

Source: Data collected during interviews with company officials.

Exhibit 10

EAGLE MANUFACTURING COMPANY

Cost of Tea Flask, June 1973*

Glass, Raw Material	2.95	
Labor	6.86	
Other direct + indirect**	8.48	
Amortization	4.12	
Total Glass		22.41
Vacuuming & Silvering,		
Raw Material	5.88	
Labor	4.02	
Other direct + indirect**	15.00	
Amortization	1.99	
Total, Vacuuming & Silvering		26.89
Non-Glass, Raw Material		36.80
Assembly, Labor	1.96	
Other direct + indirect**	2.65	
Amortization	1.39	
Total Assembly		6.00
Total Manufacturing Cost		92.10
Cost of selling, finance, administration, on the basis of 20% of manufacturing costs.		18.42
Total Costs		110.52

*According to company records, the average manufacturing cost during the fiscal year was Rls. 90.26. A breakdown of this figure was unavailable. The company produced a total of 347,242 tea flasks during the year.

**Based on standard rates of total section expense to direct labor expense.

Source: Data collected during interviews with company officials.

Exhibit 11

EAGLE MANUFACTURING COMPANY

Cost of Coleman Thermos, June 1973*

Parts and Raw Material		161.50
Assembly, Labor	17.15	
Amortization	5.49	
Other indirect	6.12	
Total		28.76
Labor, painting, metal work, insulating		29.25
Packaging, small carton, large carton, tape, plastic bag.		18.08
Total Manufacturing Costs		237.59
Admin Overhead (indirect)		28.00
Loss due to rejected & returned goods		13.28
Total cost		278.87

* According to company records, the average manufacturing costs during the fiscal year was Rls. 254.58. A breakdown of this figure was unavailable. The company produced a total of 96,879 Coleman thermoses during the year.

Source: Data collected during interviews with company officials.

Exhibit 12

EAGLE MANUFACTURING COMPANY

The Coleman Thermos

(2) Outer and inner caps are linear polyethylene, unaffected by temperature.
Outer cap locks positively won't cross theard, stick or bind, crack or split.
Inner lid is full - sized cup with handle.
No need to carry drinking utensil.

(1) Urethane insulation-the world's finest - wraps its blanket of millions of tiny gas cells to lock in cold, Seals out condensation and heat.
Will not deteriorate with age.

(3) Rugged bail handle for ease of carrying

(4) Upper casing formed of single piece, Heavy _ gauge steel with new smooth streamlined styling. No leaky seams to allow rust or corrosion Baked enamel finish.

(5) Bottom of rugged plastic is rustproof lightweight, and permanently handsome. Pioneered by Coleman, it's still the best of its kind.

(6) FAST _ Flo push _ button faucet fills glasses faster.
Empties one gallon jug in one minute.
Won't sweat or drip.
Pour spout models with new flip cap.

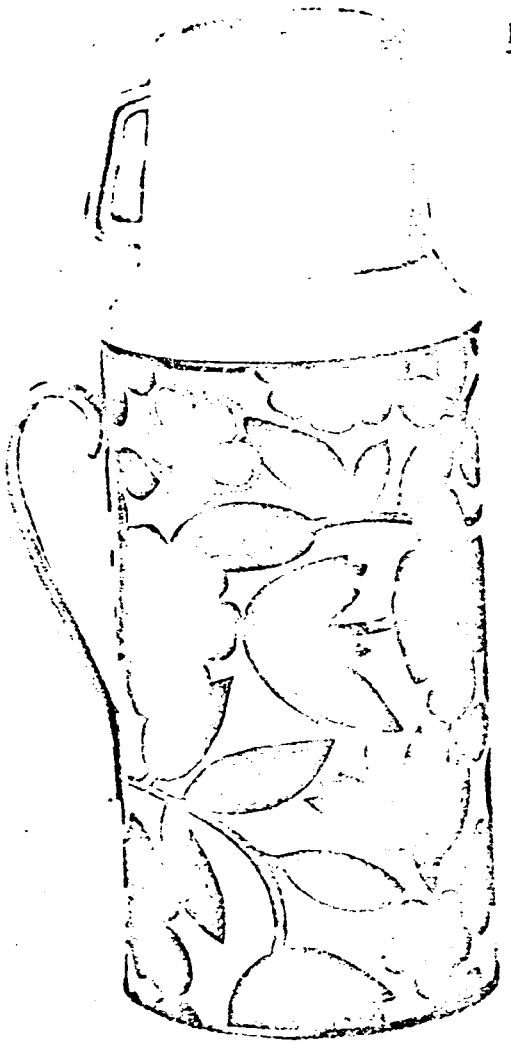
(7) One piece molded liner is stain and ordor - resistant, can't crack or chip, requires no per-cooling.

Source: EMC Sales Brochure

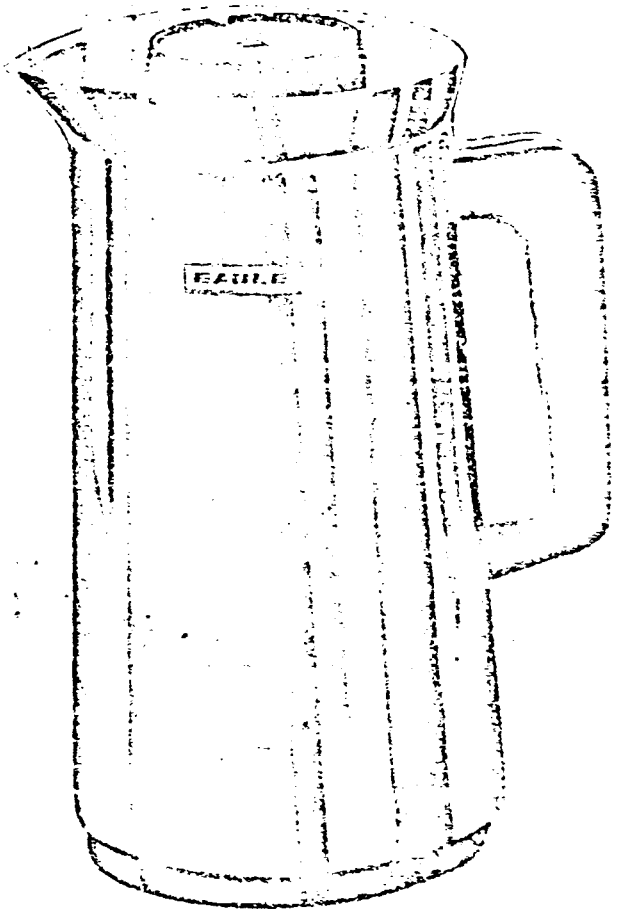
Exhibit 13

EAGLE MANUFACTURING COMPANY

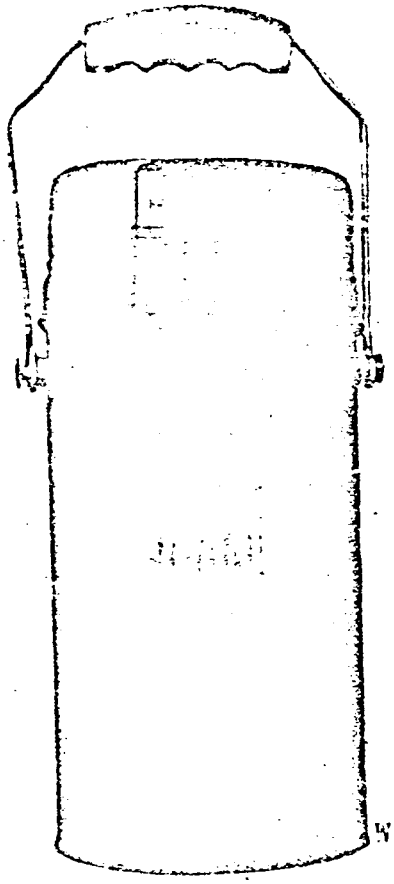
Eagle Ice Flasks and Tea Flasks



Tea Flask



Steel Tea Flask



Wide Mouthed Ice Flask

Source: EMC Sales Brochure

Table 1

EAGLE FLASK COMPANY

Imports of Tea Flasks During 1972

	Imports During the last nine months of 1972			Averages	
Source	units (000)	Kg. (000)	CIF Value Rls. (000)	Kg./unit	Rls./unit
Republic of China	155	86	7,767	0.555	50.1
Taiwan	27	15	1,728	0.556	64.0
Japan	11	11	870	1.000	79.0
Hong Kong	5	3	490	0.600	98.0
Austria	7	4	624	0.572	89.2
Italy	2	1	200	0.500	100.0
U.K.	25	14	2,312	0.560	92.5
U.S.A.	3	2	623	0.667	207.6
Other	35	19	1,576	0.544	45.0
TOTAL	270	155	18,600	0.572	68.9

Source: Ministry of Finance Statistics Department

Table 2

EAGLE FLASK COMPANY

Unit Cost Comparisons of Imported Flasks, 1972 and 1973

Source	CIF Price	Customs Duties		Commercial Tax		Transportation Costs	Cost to Wholesaler	
		1972	1973	1972	1973		1972	1973
Republic of China	50.1	22.2	20.0	33.3	39.9	2.5	108.1	112.5
Taiwan	64.0	22.3	25.6	33.4	40.0	3.2	122.9	132.8
Japan	79.0	40.0	31.6	60	72	4.0	183	186.6
UK	92.5	22.4	37.0	33.6	40.3	4.6	153.1	174.4
Other	45	21.8	18	32.6	39.2	2.3	101.7	104.5

Source: Estimated by application of the rates for customs duties and commercial taxes to published figures of Ministry of Finance statistics Department.